

# Capstone Defense Cheat Sheet

What Canary is, how each component works, what the results mean, and which claims the evidence supports

**Core story:** Identify buildings that need attention, project Day 35 weight and harvest recovery, explain the warning signs, and recommend the next management check.

Updated 12 August 2026. Use this as the shared answer key for the dashboard, notebooks, model files, and presentation.

## One-page “memorize this” summary

### What is Project Canary?

Project Canary is an early-warning and decision-support system for a broiler farm. Its purpose is to help make production outcomes more consistent by giving management earlier visibility of buildings that are going off-track. It identifies recorded warning signs, estimates harvest recovery and Day 35 average weight, and links confirmed operating patterns to Doc Raymond's action playbook.

Canary does not directly control mortality or growth. Its value is earlier visibility, so management can investigate and intervene before weak daily signals become final outcomes.

### What business question does it answer?

**How can the farm make production outcomes more consistent? Specifically: which buildings are going off-track, what evidence explains the concern, what harvest recovery and Day 35 weight are currently expected, and what should management check first?**

The Day 35 weight milestone is **1,800 g**; the harvest-recovery goal is **95%**. Day 14 is the principal early-warning checkpoint, but Canary continues updating through Day 35 and harvest.

### What are the three engines?

1. **Risk scoring:** transparent rules score four observed dimensions from 0–3 points each. Total 0–12 becomes Low, Medium, High, or Critical.
2. **Predictive outlooks:** a recovery estimator and a Day 35 weight method are evaluated independently using data available by the selected review date.
3. **Recommendation playbook:** deterministic triggers map to Doc Raymond's recorded checks, urgency, and escalation. It does not diagnose disease or automatically treat birds.

## What data do we have?

- The newest workbook contains **1,624 unique building-day records with no duplicate building-day keys**. The row count fell because unsupported forward-filled Days 36–49 were removed from 2026-3; this prevents Day 35 population from being mistaken for a later harvest endpoint.
- Recovery: **31 independent building-cycle outcomes**, represented by **151 balanced decision snapshots**; 1,355 leakage-safe daily snapshots are retained for audit.
- Day 35 weight: **31 independent building-cycle outcomes**, represented by **124 Day 7/14/21/28 checkpoint rows**.
- Recovery endpoint is the agreed proxy: last-recorded population ÷ beginning population. The redesigned **training Y** is the additional population loss after each review date.
- Weight endpoint is observed average bodyweight on Day 35. The redesigned **training Y** is the remaining gain from the current checkpoint to Day 35.
- Day 35 goal: **1,800 g**. Recovery goal: **95%**.

## What models are operational?

- **Recovery: ordinary linear remaining-loss regression** is the operational continuous-estimate model. Whole-cycle MAE is **1.74 percentage points**; cycle-balanced MAE is **1.76 points**; RMSE is **2.57 points**;  $R^2$  is **0.054**. It improves cycle-balanced MAE by **16.1%** over the refreshed age-band baseline and is effectively tied with Ridge while remaining simpler to explain. It remains experimental because at/above-95% recall is only **21.1%**.
- **Day 35 weight: historical remaining gain** remains operational. MAE is **178 g**; cycle-balanced MAE is **182 g**; RMSE is **242 g**;  $R^2$  is **0.126**; **65.3%** are within 200 g. Every learned challenger performs worse on unseen cycles.
- **Later 2026-3 audit**: weight was encouraging at **63 g MAE** across 12 checkpoint forecasts, but recovery was weak at **4.08 percentage-point MAE** and biased low. These three later buildings were excluded from all model selection. The recovery result is a clear warning that more completed cycles and a verified harvest endpoint are needed.

## What is the honest claim?

Canary is a **validated prototype**, not a production guarantee. Recovery is useful directionally as a continuous estimate, but its low held-out  $R^2$  means most across-cycle variation remains unexplained. Day 35 weight uses the more defensible transparent fallback because learned models still do not improve enough on unseen cycles.

## Five-step explanation: Engine 1 — Risk scoring

1. **Question**: Which building shows the most operational concern today?
2. **Inputs**: weight gap versus the approved age target; survival/population-loss evidence; recent mortality; and environmental evidence against age-specific temperature/humidity bands. Missing or stale evidence is disclosed.
3. **Process**: each dimension receives 0–3 points using configurable thresholds. Available points are added; the trace shows each observation, rule, score, and missing dimension.
4. **Output**: total score, Low/Medium/High/Critical label, leading problem pattern, and “why now” evidence.
5. **Limitation**: the score is an operational-priority rule, not a probability of missing 95% recovery or 1,800 g. Environmental thresholds remain subject to farm approval.

## Why rules rather than a risk classifier?

- Only 31 independent outcomes are available.
- The owner needs to see the exact rule and measurement behind every flag.
- Rules keep operating concern separate from statistical forecasts.
- Thresholds can be reviewed and changed without retraining a model.

## Five-step explanation: Engine 2A — Recovery forecast

1. **Question:** Using only what is known today, what final recovery proxy should we expect?
2. **Inputs (X):** age and remaining days; current survival and cumulative loss; mortality level/acceleration; latest weight gap/freshness; temperature/humidity exposure and freshness. Feed is withheld until its unit is confirmed.
3. **Process:** create Day 7/14/21/28/latest snapshots; give every building-cycle equal total weight; impute, scale, tune, and compare models inside nested whole-cycle folds.
4. **Output:** current survival minus predicted remaining loss, plus target gap, empirical range, model version, forecast date, and predictive context.
5. **Limitation:** Y is a last-recorded recovery proxy, only six historical cycles are available, and the model strongly recognizes below-target cases but still misses most at/above-95% cases.

## Recovery workflow — explain it from start to finish

| Stage                      | What Canary does                                                                                                                                                                                 |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data</b>                | Starts from the corrected farm workbook, consolidates Zone A/B rows, and produces one building-day record.                                                                                       |
| <b>Y output</b>            | Calculates remaining loss after the snapshot: current percentage alive – completed-cycle recovery proxy. Final forecast = current survival – predicted remaining loss.                           |
| <b>X inputs</b>            | Uses only evidence known on the review date: age, current survival, mortality signals, latest weight gap/freshness, and environmental deviations/freshness.                                      |
| <b>Feature engineering</b> | Converts raw readings into per-bird rates, gaps from approved targets, rolling mortality signals, days outside environmental bands, and staleness flags.                                         |
| <b>Pre-processing</b>      | Inside each training fold only: median-impute missing numeric values, add missingness indicators, and standardize inputs for linear models.                                                      |
| <b>Validation</b>          | Removes one complete harvest cycle at a time; the inner loop selects settings using only the remaining cycles. Repeated snapshots are weighted so each building-cycle has equal total influence. |
| <b>Selection</b>           | Compares five declared methods on cycle-balanced MAE, then checks $R^2$ , cycle stability, target-side performance, and champion gates.                                                          |
| <b>Live output</b>         | Returns current survival minus expected remaining loss, a likely range, gap to 95%, model version, and non-causal driver context.                                                                |

**Memorize: what “held-out cycle” means**

Recovery uses six complete-cycle folds: **2025-2, 2025-3, 2025-4, 2025-5, 2026-1, and 2026-2**. In each test, every building from one named cycle is removed, Canary learns from the other cycles, and then it predicts the excluded cycle as if it were new. The current 2026-3 cycle is never part of historical model selection. The weight model uses the same six historical folds; its newly observed 2026-3 weights are reserved as a small prospective audit.

This applies to predictive models. Risk alerts are transparent rules, so they are replayed using a historical review date rather than trained and held out.

**Memorize: Day 14 retrospective**

1. Freeze each completed building at Day 14.
2. Remove its entire cycle from training.
3. Estimate the additional loss after Day 14 from the other cycles.
4. Calculate **Day 14 survival – expected remaining loss**.
5. Compare with **last-recorded population ÷ beginning population**.

Example — **2025-2 Tags 1**: projected **92.75%**, recorded proxy **94.28%**, error **-1.53 points**.

Day 14 recovery results: **MAE 1.95 points, RMSE 2.71 points, bias +0.49 points, R<sup>2</sup> 0.025**, and empirical 80% range about **±2.70 points**. Target-side accuracy is **87.1%**, equal to the always-below majority baseline, with **0% recall** of actual ≥95% outcomes. Day 14 remains useful for early operational warning, but its recovery forecast is directional rather than a reliable target-hit classifier.

**Memorize: live dates between checkpoints**

The selected ordinary linear remaining-loss model can score any review date using that date's leakage-safe feature snapshot. If the fitted model is unavailable, Canary falls back to the transparent Day 7/14/21/28 remaining-loss baseline and interpolates between checkpoints so the fallback does not jump abruptly.

Revenue at risk is not another model:

**beginning birds × recovery gap to 95% × assumed sale weight × price/kg**

Therefore, recovery forecast changes directly change the displayed revenue scenario.

**Recovery Y target**

**additional loss Y = current percentage alive – final recovery proxy**

**predicted final recovery = current percentage alive – predicted additional loss**

The value is used only after prediction for evaluation. Current survival may be an X because it is already known on the review date and logically constrains what final recovery can be. Future ending population is never an X.

## Recovery model comparison — primary prospective test

| Candidate                               | MAE (pts)   | Cycle-balanced MAE (pts) | RMSE (pts)  | R <sup>2</sup> | Decision                            |
|-----------------------------------------|-------------|--------------------------|-------------|----------------|-------------------------------------|
| Age-band remaining-loss baseline        | 2.00        | 2.09                     | 2.66        | -0.010         | Transparent baseline                |
| <b>Linear remaining-loss regression</b> | <b>1.74</b> | <b>1.76</b>              | <b>2.57</b> | <b>0.054</b>   | <b>Selected continuous estimate</b> |
| Ridge remaining-loss regression         | 1.74        | 1.76                     | 2.57        | 0.055          | Compared                            |
| Gradient Boosting remaining-loss        | 1.76        | 1.84                     | 2.47        | 0.129          | Compared                            |
| Constrained Extra Trees                 | 1.73        | 1.80                     | 2.47        | 0.129          | Nonlinear sensitivity challenger    |

### Why select linear regression—and why not call it a 95% classifier?

Linear regression improves cycle-balanced MAE by **16.1%** versus the refreshed age-band baseline, keeps positive held-out R<sup>2</sup>, and is effectively tied with Ridge. Extra Trees has slightly lower pooled MAE and higher R<sup>2</sup> but worse cycle-balanced MAE. The simpler linear method therefore wins under the prespecified cycle-balanced MAE and simplicity rule. However, it detects only **21.1%** of actual at/above-95% snapshots. Canary therefore shows a point estimate and range, not a probability that the flock will hit 95%.

### Recovery classification warning

- Target-side accuracy: **90.1%** for the selected linear estimate.
- Majority baseline: **87.4%** overall.
- Day 14: **87.1%**, equal to the **87.1%** majority baseline.
- Below-95% recall: **100%**.
- At/above-95% recall: **21.1%**.
- Balanced target accuracy: **60.5%**; still operationally weak because the positive-side sample is small.

Therefore, do not say “the model accurately classifies whether recovery will hit 95%.” Say: **it estimates the continuous recovery level, but target-side classification is not validated.**

### Recovery top held-out drivers

Canary reports held-out permutation importance for the operational linear model and held-out SHAP for the strongest nonlinear sensitivity challenger, Extra Trees. SHAP is calculated only after removing the complete test cycle. It shows which inputs moved model estimates and whether higher values generally moved the estimate up or down. These are predictive associations—not proof that changing a factor will cause recovery to improve. Building-specific SHAP explains one forecast; actual actions still require a recorded rule violation and Doc Raymond's playbook.

## Five-step explanation: Engine 2B — Day 35 weight

1. **Question:** Given recorded growth so far, what average building weight should we expect on Day 35 against the 1,800 g goal?
2. **Inputs (X):** latest/checkpoint weights; weighing day; weight-to-target ratio; recent and cumulative average daily gain; earlier checkpoint weights available by that date; current survival; mortality; and environmental exposure/freshness.
3. **Process:** use only observed Day 7/14/21/28 weights; hide future checkpoints; give each building-cycle equal total influence; run nested whole-cycle validation and apply strict replacement gates.
4. **Output:** projected Day 35 weight, gram and percent gap to 1,800 g, range, method, measurement day, and staleness.
5. **Limitation:** only 31 independent Day 35 outcomes exist; 26 are below target and five are at/above target; no learned model clears every gate.

### Day 35 weight workflow — explain it from start to finish

| Stage               | What Canary does                                                                                                                                                                                                                                                        |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data                | Uses corrected, observed building weights and keeps observed values separate from target-curve interpolation.                                                                                                                                                           |
| Y output            | Calculates remaining gain: observed Day 35 weight – current checkpoint weight. Final forecast = current weight + predicted remaining gain.                                                                                                                              |
| X inputs            | At each Day 7/14/21/28 checkpoint, exposes only weights and operating evidence already recorded by that checkpoint. Later weights remain blank.                                                                                                                         |
| Feature engineering | Creates measurement age, target ratio/deficit, average daily gain, available prior checkpoint weights, survival, environmental exposure, missingness, and staleness.                                                                                                    |
| Pre-processing      | Performs imputation, missingness handling, scaling, feature filtering, and tuning inside training folds—not before the holdout.                                                                                                                                         |
| Validation          | Removes a complete cycle, trains on the other cycles, predicts the unseen cycle, and repeats. Every building-cycle has equal total influence.                                                                                                                           |
| Selection           | Compares historical remaining gain, checkpoint linear, Ridge, robust Huber, and constrained Gradient Boosting. A learned model must improve cycle-balanced MAE by 10%, keep positive $R^2$ , remain stable, reach 70% within 200 g, and improve target-side usefulness. |
| Live output         | Because no learned model passes, adds the training-only historical remaining gain for the measurement age to the latest observed building weight.                                                                                                                       |

### Day 35 weight Y target

**remaining gain Y = observed Day 35 weight – current checkpoint weight**

**projected Day 35 weight = current checkpoint weight + predicted remaining gain**

The approved target curve is a known reference used to calculate age-specific progress. It never replaces a missing observed Y.

## Weight model comparison — primary prospective test

| Candidate                        | MAE   | Cycle-balanced MAE | RMSE  | R <sup>2</sup> | Within 200 g | Decision                  |
|----------------------------------|-------|--------------------|-------|----------------|--------------|---------------------------|
| Historical remaining gain        | 178 g | 182 g              | 242 g | 0.126          | 65.3%        | Operational fallback      |
| Checkpoint linear remaining-gain | 216 g | 208 g              | 273 g | -0.118         | 56.5%        | Compared                  |
| Ridge remaining-gain             | 207 g | 200 g              | 264 g | -0.045         | 56.5%        | Compared                  |
| Robust Huber remaining-gain      | 226 g | 223 g              | 276 g | -0.144         | 51.6%        | Compared                  |
| Gradient Boosting remaining-gain | 203 g | 207 g              | 262 g | -0.026         | 52.4%        | Best nonlinear challenger |

### How historical remaining gain works

For each checkpoint age in the training cycles:

**remaining gain = observed Day 35 weight – observed checkpoint weight**

Average those gains using training cycles only, then:

**projected Day 35 weight = current observed weight + average historical remaining gain for that age**

The held-out cycle is excluded when validating, so its Day 35 result cannot influence its own prediction.

### Why not use a learned model as the live method?

All learned challengers have negative whole-cycle R<sup>2</sup> and higher cycle-balanced MAE than the historical remaining-gain baseline. The approved gate requires at least 10% improvement, positive R<sup>2</sup>, stability, and at least 70% within 200 g. Using the baseline is therefore the honest, defensible choice.

### What drives the live weight projection?

The live formula uses three direct items: latest measured weight, its measurement day, and the average historical remaining gain for that checkpoint. Learned-model importances may be shown as research evidence, but they do **not** drive the live forecast.

### Day 14 weight proof

- 31 building outcomes tested.
- Day 14 MAE: **181 g**.
- Day 14 RMSE: **237 g**.
- Within 200 g: **58.1%**.

- At/above-1,800 g recall: **0%**.

Day 14 remains the main early-warning checkpoint because it is early enough to act and its observed weight is used in both target-gap monitoring and the later-growth forecast. It is a useful signal, not proof of causality.

## Five-step explanation: Engine 3 — Recommendations

1. **Question:** What should management inspect next, given the confirmed warning pattern?
2. **Inputs:** deterministic risk/alert pattern, severity, recorded evidence, measurement freshness, and approval status.
3. **Process:** match the trigger to Doc Raymond's playbook—for example low bodyweight, high mortality, high/low temperature, high/low humidity, abnormal fluctuation, low feed intake, or poor recovery outlook.
4. **Output:** plain-language check, urgency, escalation condition, and applied rule.
5. **Limitation:** recommendations are inspection/management guidance. They do not diagnose disease, estimate causal treatment effects, or prescribe automatic treatment.

## Why nested whole-cycle validation?

- **Outer loop:** remove every row from one harvest cycle; train and tune without it; predict it; repeat for all cycles.
- **Inner loop:** choose hyperparameters using only the remaining training cycles.
- Imputation, scaling, and tuning never see the outer test cycle.
- Repeated snapshots are weighted so one building-cycle has the same total influence as another.

This answers the deployment-like question: **can Canary generalize to a future unseen harvest cycle?**

The colleague's building-cycle LOGO test answers an easier question: can it predict one building when other buildings from the same cycle remain in training? Its higher  $R^2$  is not directly comparable.

## What we learned from the teammate model

### Incorporated

- A structured and reproducible cleaning pipeline.
- Zone A/B aggregation before feature creation.
- Environmental, growth, missingness, and freshness features.
- Group-aware cross-validation and comparison with boosted trees.
- Explicit model comparison and feature-importance reporting.

### Not adopted as the primary Canary test

- Treating repeated daily rows as new independent flock outcomes.
- Leaving out one building while keeping other buildings from the same cycle in training.
- Using interpolated early weights that were created with later observations.
- Selecting features on the full dataset before cross-validation.
- Using a pickle or headline score that cannot be reproduced from the corrected canonical workbook.

**Defense answer:** We took the useful engineering ideas, but retained stricter whole-cycle validation because Canary must generalize to a future cycle, not merely another building in a cycle it has already seen.



## Why no SMOTE or oversampling?

- These are regression problems, while standard SMOTE is primarily for classification.
- The scarce evidence is independent building-cycles, not spreadsheet rows.
- Synthetic rows do not create new cycles and can make uncertainty look falsely small.
- Synthetic poultry trajectories may be biologically implausible.

## How to interpret MAE, RMSE, $R^2$ , and bias

- **MAE:** typical absolute miss in business units; primary selection metric.
- **RMSE:** penalizes large misses more strongly.
- **$R^2$ :** fraction of variance explained on held-out data; useful diagnostic, not the sole decision rule.
- **Bias:** whether predictions are high or low on average.
- **Target-side accuracy/recall:** whether a model distinguishes the two sides of 95% or 1,800 g; always compare with the majority baseline.

Low  $R^2$  means much variation remains unexplained. It does not erase all usefulness, but it requires prototype positioning, wider uncertainty, and conservative claims.

## Seven EDA questions Canary answers

1. **Coverage:** Which cycles, buildings, days, and measures are complete enough to analyze?
2. **Early growth:** Is Day 14 weight associated with Day 35 weight?
3. **Early recovery signal:** Is Day 14 weight associated with the final recovery proxy?
4. **Environment:** Are recorded temperature and humidity within the provisional age-specific bands, and is the evidence balanced enough to compare outcomes?
5. **Survival paths:** When do buildings begin to drift below the expected survival path?
6. **Model accuracy:** How large are forecast errors overall, by checkpoint, and by held-out cycle?
7. **Target attainment:** How often have completed building-cycles reached 1,800 g, 95% recovery, or both?

**Defense answer:** EDA is used to test the business story, expose data limitations, inform transparent features, and prevent overclaiming. It does not prove that changing one factor will cause a better outcome.

## Common panel questions

### “Did you train on daily rows as if they were independent flocks?”

No. The independent outcome is one building-cycle. Repeated snapshots give multiple historical decision points, but they receive equal building-cycle weighting and are always held out together by cycle.

### “Did you leak future weights?”

No. At a Day 14 snapshot, Day 21/28/35 inputs are blank. The Day 35 value is used only as Y after prediction. Target-curve interpolation is a known reference, not an observed weight.

**“Why is current survival allowed in recovery X?”**

It is known today and directly limits what final recovery can be under the agreed formula. Future ending population is never used as X. Raw beginning population is excluded unless the farm later approves a meaningful density feature.

**“Can feature importance tell the owner to lower temperature by 3°C?”**

No. Importance shows association and predictive reliance. The action table must be triggered by an observed threshold and approved farm guidance, not by a causal claim from observational data.

**“Are these models reliable?”**

They are useful as validated prototype estimates, not guarantees. The linear recovery model reduces typical error but has low  $R^2$  and weak at/above-95% recall. Weight learned models fail the replacement gates, so Canary uses historical remaining gain.

**“What would most improve the models?”**

Collect more completed cycles with verified harvest dates/populations, consistent bodyweight sampling, confirmed feed units, complete environment readings, and more outcomes on both sides of the targets.

**Where to show proof in the deliverables**

- **Canary Methodology page:** five-model tables, gates, timing and cycle performance, drivers, and limitations.
- **Building View:** current evidence, forecast trace, and building-specific linear contribution view where available.
- **EDA & Insights:** data coverage, early-weight relationships, target attainment, and limitations.
- **outputs/model\_ready/Project\_Canary\_Model\_Ready\_Data.xlsx:** outcomes, exact training rows, data dictionary, and daily audit.
- **notebooks/Project\_Canary\_Harvest\_Recovery\_Model.ipynb**
- **notebooks/Project\_Canary\_Day35\_Weight\_Model.ipynb**
- **models/harvest\_recovery\_champion.pkl**
- **models/day35\_weight\_champion.pkl**

**Outstanding farm validation**

1. Approve environmental thresholds and interventions.
2. Confirm feed-intake units and targets.
3. Define a consistent bodyweight sampling procedure.
4. Confirm the eventual verified harvest-date and ending-population source.
5. Define acceptable forecast-error limits for management use.